

Indian Institute of Information Technology Bhagalpur 813210

Department of Electronics & Communication Engineering
End Semester, April 2023

B. Tech. 2nd Year- ECED & CSED, Semester IV

Subject: EC208: Microprocessor & Interfacings

Date: April 24, 2023

Duration: 180 Minutes

Maximum Marks: 50

Instructions: Assume necessary data wherever required.
Draw appropriate figure wherever necessary.
Figures in the right margin indicate marks.

1. ☒ a) Given that $BX = 2500H$, $SI = 5000H$, Displacement = $1000H$, $IP = 2000H$ [3]
Determine the effective address for the following addressing modes:
i) Immediate
ii) Base Index
iii) Index
- ☒ b) Explain in detail the coding templates for MOV instruction of 8086. [2]
- ☒ c) Draw the block diagram of 8086 and explain the functions of general-purpose registers (GPRs). [4]
- ☒ d) How many bytes can be stored in the queue of 8086? [1]
2. ☒ a) While executing a program, when the 8085 MPU completes the fetching of the machine code located at the memory address $2057H$, what is content of the program counter? [1]
- ☒ b) i) Write an assembly language program to convert a given 8-bit binary number into a Gray code. [4]
ii) Write an assembly language program to find two's complement of an 8-bit number.
- ☒ c) Draw the flag register of 8086. [1]
- ☒ d) Write a program to use Counter 1 of 8254 in mode 3 as a square wave generator. Assume $N = 12$. [4]
The counter operates as a binary counter.
3. ☒ a) The two machine codes - 00111110 ($3EH$) and 00110010 ($32H$) are stored at memory locations $2000H$ and $2001H$ respectively. The first machine code ($3EH$) represents the opcode, and the second code ($32H$) is the data to be loaded into the accumulator. If the clock frequency is 2 MHz, draw the timing diagram and calculate the time required to execute the opcode fetch and memory read cycles. [4]
- ☒ b) Which steps 8086 follows to handle any interrupt. What do you mean by NMI interrupt? [3]
- ☒ c) Illustrate the data flow and list the sequence of events when 1- byte instruction is fetched by the microprocessor unit. [3]
4. ☒ a) What are the two modes of 8086? Draw the pin diagram of the 8086 micro-processor. [2]
- ☒ b) Design a seven-segment LED output port with the device address $F5H$, using a 74LS138 3-to-8 decoder, a 74LS20 4-input NAND gate, a 74LS02 NOR gate, and a common-anode seven-segment LED to display letter "7". [4]
- ☒ c) Explain Programmable Peripheral Interface (PPI) 8255 with suitable block diagram. What should be the control word if Port A and Port B are to be configured as input and Port C_{lower} and Port C_{upper} have to be configured as output ports. Assume mode to be Mode 0. [4]

5. a) Design an 8-bit DAC0800 and obtain the resistance ratio of MSB and LSB. [5]
- b) Explain the architecture of 8087 with the help of suitable diagram? [5]

Name:

Roll No.:

*****Wish you all best*****